

- ## 09 Executive BI & Mobile Analytics

The **PCode Generation & Inventory Master Module** serves as the technological core of the entire ERP suite. It governs how parts are onboarded, verified, secured, priced, and substituted across all branches.

1. Multi-Company & Branch Matrix

Creation of company entities, country assignments (Bahrain/Dubai), base currency configs (BHD/AED), VAT parameters, and branch categorization (Stores vs Outlets).

2. Sequential PCode FIFO Engine

Automated, collision-proof sequence generation, real-time uniqueness validation, item-wise save states (Red=Unsaved, Green=Saved), and permanent lock upon CTO approval.

3. Comprehensive Item Master

20+ core attributes including multi-OEM cross-indexing, manufacturer part numbers, vehicle compatibility, engine CC/specs, multi-image assets, and tax-inclusive pricing.

4. Dual Approval Workflows

Structured approval pipelines for: (a) Sales Counter Request via Branch Purchase Manager (BPM) → CTO; and (b) Vendor Receiving variance via Store Purchase Manager (SPM) → CTO.

5. Non-PCode Exception Holding

Dedicated holding buffer for emergency customer orders, linked to pending CTO review dashboard with automated sequential migration upon formal approval.

6. Centralized Pricing Hub

Single-window country-wise price updates, automated exchange rate calculations, bulk updating, role-based modification locks, and audit history versioning.

7. Minimum Order Level & Alerts

Branch/store-specific safety stock thresholds, automatic low-stock trigger engine, and consolidated executive review portal for Purchase Team and CTO.

8. Substitute & Mapping Intelligence

Automated OEM compatibility engine, multi-brand alternative suggestions, cross-tier price recommendations, and confidential trade-secret protection.

9. Catalog Sync & Sales Alerts

PartSouq catalogue integration (API/Sandbox), item handling precautionary sales pop-ups (sticker/box removal), legacy PCode archiving, and CTO analytics audit screen.

03 Project Team Structure & Resource Allocation Matrix

To guarantee timely execution, highest engineering quality, and seamless business alignment, SaNDS Lab Middle East W.L.L will deploy a dedicated multi-disciplinary delivery team consisting of experienced Project Managers, Software Architects, Senior Developers, and Quality Assurance Specialists.

Role	Key Responsibilities	Allocated Team	Total Effort	Focus Area
Project Manager (PM) & BA	Sprint governance, milestone tracking, client communication, UAT coordination, requirement sign-off.	1 LEAD PM	160 Hours (10 Wks)	Governance & Delivery Control
Back end Lead Eng	REST API development, database schemas, PCode sequential engine, approval workflows, caching, audit trails.	1 SENIOR BACKEND DEV	280 Hours (10 Wks)	Core Business Logic & APIs
Front End Lead Eng	Responsive UI/UX design, real-time validation grids, Item Master interface, CTO approval dashboard, sales alerts.	1 SENIOR FRONTEND DEV	240 Hours (10 Wks)	User Experience & Rapid Entry
Database and Cloud (DevOps)	Multi-tenant DB clustering, backup automation, PartSouq API connector, CI/CD pipelines, security hardening.	1 DEVOPS SPECIALIST	120 Hours (10 Wks)	Cloud Infra & Data Security
QA (Testing & Verification)	Functional test suite, edge-case validation, concurrency tests on PCode generation, security audit, UAT scripts.	1 QA LEAD / TESTER	180 Hours (10 Wks)	Defect Zero & Rigorous QA
Total Dedicated Engineering & Governance Effort			980 Person-Hours (10 Calendar Weeks)	

04 Detailed Milestone Breakdown & Deliverables

The implementation of Module 1 is structured into **six (6) clear, verifiable milestones** spanning 10 working weeks (50 active business working days, excluding Bahrain weekends and official public holidays). Each milestone culminates in demonstrable software artifacts, client reviews, and formalized sign-off criteria.

**MILESTONE
01**

**Architecture, Multi-Company Setup & Core
Security Framework**

Weeks 1 – 2 (10
Working Days)

20% • BD
681.818

TECHNICAL DELIVERABLES & KEY TASKS

- Cloud infrastructure provisioning (AWS/Azure) with high-availability database cluster and staging environment.
- Database Schema Design for Multi-Company (Bahrain HO, Dubai branch) and Branch Outlets/Stores.
- Base currency (BHD, AED, USD) exchange matrix and country-specific VAT/tax structure definition.
- Role-Based Access Control (RBAC) engine establishing permissions for CTO, BPM, SPM, Sales, and Accounts.
- Comprehensive Master Data dictionary & API Swagger documentation baseline.

ASSIGNED TEAM

Project Manager

Lead Backend Dev

DevOps Specialist

QA Engineer

Milestone Sign-off Criteria

Working cloud staging environment, verified multi-branch database hierarchy, authenticated login with role permissions demonstrated to client.

**MILESTONE
02**

**PCode Sequential Engine, Item Master & Non-
PCode Holding System**

Weeks 3 – 4 (10
Working Days)

20% • BD
681.818

TECHNICAL DELIVERABLES & KEY TASKS

- Automated Sequential FIFO PCode Generator with zero collision risk and instant uniqueness verification.
- Interactive Item Master UI with 20+ fields (OEMs, Part Numbers, Brand, Vehicle Model, CC, Fitment, Units, Tax).
- Real-time visual entry indicators: Unsaved row (Red) vs. Saved row (Green) to prevent operational data loss.
- PCode Permanent Lock mechanism triggering post-creation to ensure immutable audit integrity.
- Non-PCode Item Master holding area for emergency counter sales, tracking unregistered items with transaction links.

ASSIGNED TEAM

Lead Frontend Dev

Lead Backend Dev

QA Engineer

Milestone Sign-off Criteria

Live test showing automated PCode sequential generation, Item Master creation with images, Red/Green save indicators, and Non-PCode emergency booking.

**MILESTONE
03****Dual Approval Workflows, Centralized Pricing & Stock Alert Engine**

Weeks 5 – 6 (10 Working Days)

20% • BD
681.818**TECHNICAL DELIVERABLES & KEY TASKS**

- **Workflow Case 1 (Sales Counter):** Sales Request → Branch Purchase Manager (BPM) duplicate verification → CTO technical validation & PCode creation.
- **Workflow Case 2 (Vendor Receiving):** Store Purchase Manager (SPM) PO vs. physical variance → Brand divergence notification → CTO review & generation.
- Centralized Multi-Country Price Management single-screen interface with automated currency conversion and country overrides.
- Minimum Order Level (MOL) configuration by Store/Branch with automated dashboard triggers and Purchase/CTO review console.
- Full audit trail logging every price modification, date, user ID, and justification notes.

ASSIGNED TEAM

Project Manager

Lead Backend Dev

Lead Frontend Dev

QA Tester

Milestone Sign-off Criteria

End-to-end execution of both approval workflows across roles, successful centralized price update reflecting in Bahrain and Dubai stores, low-stock alerts firing.

**MILESTONE
04****Item Mapping, Substitutes & PartSouq Integration**

Weeks 7 – 8 (10 Working Days)

20% • BD
681.818**TECHNICAL DELIVERABLES & KEY TASKS**

- Dedicated Item Mapping Window replacing legacy text comments with structured multi-OEM relational records.
- Advanced Substitute Suggestion Engine offering sales staff compatible parts across brands and pricing tiers.
- External Catalog Reference Integration: PartSouq.com connector (API / Embedded browser sandbox) preserving in-ERP workflow.
- Item Handling Precautionary Sales Warning pop-ups (e.g., "Remove sticker", "Check fitment") with mandatory salesperson acknowledgement.
- Trade secret confidentiality enforcement: Masking internal supplier OEM data on commercial counter documents.

ASSIGNED TEAM

Lead Frontend Dev

Lead Backend Dev

DevOps Specialist

QA Tester

Milestone Sign-off Criteria

Working substitute engine suggesting alternate parts during search, successful PartSouq catalog lookup within ERP, sales alert confirmation modal verified.

**MILESTONE
05****Legacy Data Migration, Governance & CTO Executive Dashboard**

Week 9 (5 Working Days)

10% • BD
340.909**TECHNICAL DELIVERABLES & KEY TASKS**

- Legacy PCode data cleansing and migration utility from existing BUSY ERP and Excel spreadsheets.
- Archiving and inactive PCode hiding logic (excluding obsolete parts from active searches while preserving audit trails).
- Executive CTO Audit & Analytics Dashboard: Branch-wise stock visibility, price movements, purchase velocity, and demand analytics.
- System-wide performance indexing and query optimization for 100,000+ automotive spare part records.

ASSIGNED TEAM

Project Manager

Lead Backend Dev

Database Admin

Lead Frontend Dev

Milestone Sign-off Criteria

Sample legacy dataset migrated and verified, inactive PCodes successfully hidden from search, CTO dashboard presenting live analytical charts.

TECHNICAL DELIVERABLES & KEY TASKS

- End-to-end Integration and Concurrency Testing across Bahrain & Dubai simulated branch network.
- Formal User Acceptance Testing (UAT) execution with Popular Auto Spare management, CTO, and Branch Managers.
- Comprehensive User Manuals, Video Walkthroughs, and role-based training sessions for Store Supervisors and Sales staff.
- Production cutover, final live cloud deployment, SSL certificate hardening, and automated daily backup routines.
- 30-Day dedicated Post-Go-Live Hypercare and warranty support.

ASSIGNED TEAM

Full Project Team (PM, Developers, DevOps, QA Lead)

Milestone Sign-off Criteria

Formal UAT Sign-off certificate signed by Client & Consultant, system deployed to live production URL, training completed for all branches.

05 Milestone Payment Structure & Invoicing Schedule

To ensure total financial transparency and accountability, project billing is directly tethered to verifiable milestone completions and client sign-offs.

Billing Stage	Deliverable Focus / Invoicing Trigger	Timeline	Payment Share	Amount (BHD)
ADVANCE TOKEN	Whole Project Kickoff Advance (Master ERP Contract Signing)	At Signing	MASTER ADVANCE	BD 5,000.000
MS-01	Architecture, Multi-Company DB & Security Setup	Weeks 1 – 2	20%	BD 681.818
MS-02	PCode Sequence Engine & Item Master UI	Weeks 3 – 4	20%	BD 681.818
MS-03	Dual Approval Workflows & Centralized Pricing	Weeks 5 – 6	20%	BD 681.818
MS-04	Item Mapping, Substitutes & PartSouq Integration	Weeks 7 – 8	20%	BD 681.818
MS-05	Legacy Data Migration & CTO Analytics Dashboard	Week 9	10%	BD 340.909
MS-06	QA Testing, Staff Training & Live Go-Live	Week 10	10%	BD 340.910
TOTAL MODULE 1 DEVELOPMENT COST (EXCL. ADVANCE)			100%	BD 3,409.091
TOTAL INITIAL AGREEMENT VALUE (Advance Token + Module 1)			—	BD 8,409.091

Commercial & Billing Terms

- **Master Project Token:** BD 5,000.000 advance is due upon contract execution for whole ERP mobilization.
- Module milestone invoices are issued upon formal deliverable sign-off.
- Standard payment term: Within 10 business days from invoice presentation.
- All quoted prices are exclusive of statutory VAT (applied as per Bahrain tax regulations).

Server & Hardware Policy

- **Client Provisioning:** All server hosting (cloud/on-premise), POS hardware, barcode scanners, and store terminals must be provided directly by the Client.
- **Advance for Purchasing:** If SaNDS Lab is requested to procure hardware or cloud subscriptions, the **Client must provide 100% of the hardware cost in advance.**

Warranty & Hypercare Support

- Includes **30 calendar days of free post-launch Hypercare support** covering bug fixes and minor operational adjustments.
- Dedicated on-site & remote support team based in Salmabad, Bahrain.

To safeguard the commercial, operational, and legal interests of both **Popular Auto Spare & A/C Parts Co. W.L.L** (Client) and **SaNDS Lab Middle East W.L.L** (Service Provider), the following legally binding provisions govern working hours, delivery schedules, unforeseen events, and dispute resolution:

1. Working Calendar & Official Bahrain Public Holidays

- **Standard Work Week:** Sunday through Thursday (5 business days/week). All **Fridays and Saturdays are considered official non-working corporate weekend holidays**.
- **Government Public Holidays:** All official public and national holidays declared/published by the **Government of the Kingdom of Bahrain for the corporate/private sector** (e.g., National Day, Eid Al-Fitr, Eid Al-Adha, Islamic New Year, Ashura, Labour Day, Prophet's Birthday) shall be observed as non-working holidays.
- **Schedule Adjustment:** The 10-week timeline is computed based on **50 actual working days**. Target milestone delivery dates will automatically adjust and extend based on the official Bahrain national holiday calendar.

2. Force Majeure & Unforeseen Events

- **Excused Delay Events:** Neither party shall be held liable or penalized for delays caused by extraordinary events beyond reasonable human control, including **Natural Calamities / Disasters (floods, earthquakes, severe weather), Acts of God, Wars, Armed Regional Conflicts, Civil Unrest, Government-mandated Curfews/Lockdowns, or Nationwide Telecommunication / Undersea Cable Disruptions**.
- **Automatic Extension:** In any Force Majeure event, the affected project milestone timeline shall automatically receive a **day-for-day extension** without financial penalties or liquidated damages.

3. Late Delivery & Liquidated Damages (SLA Penalty)

- **On-Time Delivery Guarantee:** SaNDS Lab guarantees milestone delivery within the agreed schedule (subject to timely client reviews, data inputs, and working day calendar).
- **Delay Penalty (Liquidated Damages):** In the event of an unexcused delay caused solely by the Service Provider exceeding a **15-calendar-day grace period**, a penalty of **0.5% per week of delay** (calculated on the affected milestone value) will be credited to the Client, capped at a maximum of **5% of that specific milestone value**.
- **Hardware & Vendor Delay Exclusion:** Any delay in the procurement, dispatch, international shipment, customs clearance, or vendor delivery of servers, cloud subscriptions, or physical hardware (e.g., POS terminals, barcode scanners) shall **NOT be considered a project delay or breach of SLA by SaNDS Lab**. An automatic day-for-day timeline extension will apply.
- **Other Exclusions:** Delays resulting from client-side approvals, third-party API dependencies (e.g., PartSouq downtime), client infrastructure delays, or approved Change Requests (CR) shall grant an automatic schedule extension without penalty.

4. Non-Payment, Default & Work Suspension

- **Payment Window:** Invoices must be settled within **10 business days** of formal milestone acceptance.
- **Right to Suspend Work:** If any milestone invoice remains unpaid past **15 calendar days** following written notice, SaNDS Lab reserves the legal right to pause subsequent milestone development and staging access without liability for project timeline slippage.
- **Reactivation:** Development sprints will resume within 3 business days following complete settlement of outstanding invoices.

5. Dispute Resolution, Governing Law & Legal Litigation Jurisdiction KINGDOM OF BAHRAIN JURISDICTION

Step 1 (Amicable Resolution): In the event of any technical, operational, or commercial dispute arising from or related to this contract, senior executive representatives (CEO of SaNDS Lab and Managing Director/CTO of Popular Auto Spare) shall meet within **14 calendar days** of written notification to reach an amicable resolution in good faith.

Step 2 (Mediation & Expert Determination): If unresolved, the dispute may be referred to an independent IT Consultant (e.g. UniGlobal Consultants) for binding technical determination.

Step 3 (Legal Litigation & Court Jurisdiction): In the event of legal proceedings, this agreement shall be governed, construed, and enforced in accordance with the commercial and civil laws of the **Kingdom of Bahrain**. Both parties irrevocably submit to the exclusive jurisdiction of the **Courts of the Kingdom of Bahrain / Bahrain Chamber for Dispute Resolution (BCDR)** for any litigation, claim, or enforcement.

07 Stakeholder Authorization & Sign-Off

By signing below, the authorized representatives of **SaNDS Lab Middle East W.L.L**, **UniGlobal Consultant**, and **Popular Auto Spare & A/C Parts Co. W.L.L** agree to the milestone schedule, resource commitments, payment structure, SLA delivery guarantees, working calendar & holiday rules, Force Majeure provisions, and legal terms specified in this document.



SERVICE PROVIDER

SaNDS Lab Middle East W.L.L

Name: **Ajit Kumar KV**

Title: **CEO & Managing Director**

Date: **21-Sep-2026**

Authorized Signature & Seal



CONSULTANT & ARCHITECT

UniGlobal Consultants

Name: **Lead ERP Consultant**

Title: **Principal Business Analyst**

Date: ____-____-2026

Review & Recommendation
Signature



CLIENT REPRESENTATIVE

Popular Auto Spare & A/C Parts Co.

Name: **Managing Director / CTO**

Title: **Executive Director**

Date: ____-____-2026

Client Acceptance & Approval
Signature